

A language model is a machine that guesses the next word, one token at a time. This wall shows every guess it made, how sure it was, and what it almost said instead.

IIAT HACLAB

GONZAGA UNIVERSITY · SEPTEMBER 2026

## The box in the room

## NVIDIA L40

ONE GPU · 46 GB MEMORY

## Qwen3 8B

## THE MODEL · 8 BILLION WEIGHTS

## 36 layers

EACH TOKEN PASSES THROUGH ALL OF THEM

**~30 / s**

TOKENS GENERATED PER SECOND



## Your sentence becomes tokens

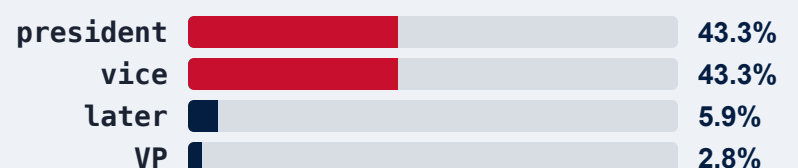
## WHAT THE MODEL ACTUALLY READ

How many r 's are in strawberry ?

**“strawberry” is one token.** It has never seen the letters inside it.  
When it answers “3 r’s” in 2.4 seconds, that is memory, not counting.

## Every token is a bet

A REAL FORK • “EVERY US PRESIDENT IN ORDER...”



A dead heat. Out of the box the wall takes the top one (**greedy**, temperature 0), so the same question gives the same answer twice. Turn the dials in the wall's **temperature** chip and it starts rolling dice instead.

## The answer, colored by confidence

“MONTY HALL, BUT THE HOST OPENS A DOOR AT RANDOM AND IT HAPPENS TO BE A GOAT. SHOULD I SWITCH?”

No, switching doesn't help in this case. The original Monty Hall problem assumes the host always opens a door with a goat, which gives you information. ...

■ sure · 90% and up   ■ unsure · 50–90%   ■ coin-flip · under 50%

Nothing here fell to a coin flip, but look where the doubt sits: the verdict **“No”** at **73%** and the hedges **help · in · always** at 61–65%. The facts about the puzzle run at 98–100%. It is sure of the story and less sure of the call.

## Inside the head, layer by layer

L1

L21 "yes" 14%

L26 75%

L31 decided

L36 99.98%

**“No”**

the 8B model on the wall · 72.7%

# DISAGREE

**“Yes”**

its 4B sibling · decided at layer 31 of 36 · 0.51 s

Same family, same question, opposite answers. The bigger model is right: when the host opens a door at random, switching is a coin flip. The smaller one settles on the famous answer and never looks back.

## Why it sometimes loops

Alright, Alright, Alright, Alright, Alright, Alright, Alright, Alright, Alright ...  $\times 10,000$  tokens  $\cdot$  281 seconds

temperature 0 · greedy · repeatable 0.6 · rolls weighted dice · escapes the rut

Four live dials on the wall's **temperature** chip change how the next token is picked. **Temperature** lets the chooser gamble on a runner-up now and then: that is “creativity”, and also “wrong more often”. **Top-p** and **top-k** shorten the list it may pick from. **Repeat penalty** taxes tokens it has already said, which is the direct cure for the loop above. The **leash** (max tokens) is the last brake: the server stops the answer at N tokens no matter what.